Sleep Spindles Phase-Lock to Respiration at Their Amplitude Peak, Not Onset, in Human NREM Sleep

A corrected permutation reanalysis of a companion study's full spindle catalog recovers a robust respiratory coupling missed by an onset-referenced test, with a fragile cardiac counterpart and a weak secondary link to spindle-locked geometry

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Abstract A companion three-dataset replication study (RVC_SLEEP) reported a null result for spindle-**onset** cardiac/respiratory phase-locking in human NREM sleep, using a two-stage hierarchical Rayleigh test on a 190-event, geometry-valid RichSleep subsample. An independent theoretical review of that result flagged the subsample as underpowered and recommended a pseudo-replication-safe permutation reanalysis on RichSleep's full raw spindle catalog rather than the geometry-filtered subsample. We reanalyzed all 23,029 detected RichSleep spindles (24 subjects) using a subject-specific random-phase-rotation permutation null and a first+second-harmonic omnibus statistic. Spindle-**onset** phase-locking remained null for both cardiac and respiratory channels at roughly $120\times$ the original sample size (all BH-FDR $q \geq 0.48$). However, spindle-**peak** (amplitude-envelope-maximum) phase showed a robust, population-level respiratory coupling (pooled $R_1 = 0.045$, permutation $p = 0.0005$; second-order $R = 0.665$ across 24 subjects, $p = 7 \times 10^{-6}$) and a weaker, less robust cardiac coupling (pooled $R_1 = 0.016$, $p = 0.02$; second-order $R = 0.462$, $p = 0.005$). Four independent robustness checks - duration-tercile stratification, leave-one-subject-out jackknife, deterministic split-half replication, and phase-convention calibration against the export's documented convention - confirmed the respiratory effect holds in every subsample tested, while the cardiac effect is duration-dependent and fails to replicate in one of two subject-split halves. We then asked whether alignment to this population-preferred respiratory phase (estimated on a held-out 99.2% of the catalog) moderates the size of the companion study's spindle-locked geometry change (the same 190 geometry-valid events): 2 of 18 pre-specified WLS-cluster tests survived BH-FDR (`d_l` endpoint, event-vs-pre-spindle-baseline mode, $q = 0.005$; `d` endpoint, same mode, $q = 0.021$), and a free circular-regression sensitivity check found the implied geometry-optimal phase within $11°$ of the independently-estimated timing-preferred phase for `d_l`. We conclude that respiratory phase locks to spindle peak, not onset, in human NREM sleep - resolving a design-driven false negative in the companion study - and report a fragile secondary signal that this coupling may also weakly moderate the spindle's own geometric consequence, framed here as an event-**timing** (not yet a confirmed event-**efficacy**) result for the sleep-EAP hypothesis, pending independent replication.

Author Summary

A companion study found no evidence that sleep spindles occur at a preferred moment of the breathing or heartbeat cycle, using a small subsample of the available recordings. Re-running the same test with a corrected statistical method on the full 23,029-spindle catalog, we found that while the very **start** of a spindle indeed shows no preferred phase, the moment of its **strongest** activity does: spindles tend to peak roughly a third of the way toward full inhalation, consistently across 24 sleeping adults and several independent checks. A weaker, less consistent link to the heartbeat cycle was also found. We further found a fragile hint that how closely a spindle's peak matches this preferred breathing phase relates to the size of that spindle's effect on brain-state geometry, though this needs independent replication.

1. Introduction

A companion three-dataset study, **RVC_SLEEP** (Langell, 2026), replicated a spindle-locked state-space reconfiguration (a rise in the `d_n` geometry endpoint) across three independent overnight-PSG datasets, then tested whether pre-event embodied-anchor (EAP) state moderates that effect. Its most mechanistically direct test (R6b,

"Mål B") asked whether spindle **onset** preferentially occurs at a particular cardiac or respiratory phase, using RichSleep's phase-resolved export (`event_phase_v2`). That test was null at every level tested: 0/48 within-subject Rayleigh tests and 0/2 population-level second-order Rayleigh tests, on a 190-event, geometry-valid, N2-staged subsample.

An independent theoretical review of that result (internal document, `articles/Rythmic_Variance_Control_in_Sleep_theory/plan/02_response.md`) identified two problems with treating that null as conclusive. First, the 190-event subsample discards 99% of RichSleep's spindle catalog (23,029 raw detected events across the same 24 subjects) for reasons unrelated to the phase-locking question itself - the subsample requires N2-staging **and** geometry validity, neither of which the phase-locking hypothesis needs. Second, the review's own worked critique of an earlier **proposed** correction (shuffling subject labels on the pooled angle set) noted that such a shuffle leaves the pooled circular-uniformity statistic invariant and therefore does not construct a valid null at all; it instead recommended a subject-specific random-phase-rotation surrogate, among other pseudo-replication-safe alternatives. It further noted that two external, non-NDT PSG studies - Ghibaudo, Juventin, Buonviso et al. (Ghibaudo et al., 2024) and Schreiner, Petzka, Staudigl, Staresina (Schreiner et al., 2023) - had already established population-level respiration-spindle phase-locking in independent human samples using a pooled, all-events design rather than the hierarchical per-subject-then-second-order design RVC_SLEEP used, raising the possibility that the RichSleep null was a design/power artifact rather than a genuine absence of coupling.

Core question

Does a corrected, pseudo-replication-safe permutation reanalysis of RichSleep's full spindle catalog recover population-level cardiac/respiratory phase-locking that the companion study's underpowered, two-stage design missed - and if so, at spindle onset, or at some other reference point in the event?

This article reports that reanalysis. It finds that onset-phase locking remains null even at full statistical power, but that spindle **peak** phase - the moment of maximal amplitude within the detected event, not its start - shows a robust population-level respiratory coupling and a more fragile cardiac one. Because a **timing** result (does phase predict **when** an event occurs, or here, what phase an event's peak lands at) is conceptually distinct from an **efficacy** result (does phase alignment change the **consequence** of an event that has already occurred), we additionally report a new, secondary analysis testing whether alignment to the population-preferred respiratory phase moderates the size of RVC_SLEEP's own spindle-locked geometry effect on the same 190 geometry-valid events used there.

2. Theory

2.1. RVC/EAP tiering and the timing/efficacy distinction

Within Noetic Diffusion Theory (NDT), sleep spindles are treated as candidate **Rhythmic Variance Control** (RVC) events - endogenous, TRN-implemented, rhythm-defined windows of transient gating-precision change (Langell, 2025). The **Embodied Anchoring Principle** (EAP) is explicitly tiered as a secondary, Tier-2 extension of this core mechanism, proposing that peripheral bodily rhythms (cardiac, respiratory) can additionally condition the same latent landscape - "intended for a later wave of interoception-focused… studies," per the theory document's own framing, not a claim on equal evidentiary footing with the Tier-1 RVC core.

The reviewed reformulation (`02_response.md`) sharpens EAP's sleep-specific form into two separable, non-exclusive pathways, which this article adopts as its organizing distinction:

- **Event-timing pathway.** Does body phase help determine **when** an endogenous RVC event (here, a spindle, or a specific feature of it such as its amplitude peak) occurs, i.e. is event phase non-uniformly distributed relative to the cardiac/respiratory cycle?
- **Event-efficacy pathway.** Given that an event has occurred, does the body's phase at that moment change **how strongly** the event reconfigures state-space geometry - i.e. does phase-alignment moderate the event's own downstream consequence?

R2-R5 below test the timing pathway (replacing RVC_SLEEP's underpowered onset-referenced version with a corrected, full-catalog reanalysis); R6 tests the efficacy pathway directly, using RVC_SLEEP's own geometry-valid subsample.

2.2. Corrected permutation-null construction

The statistic used throughout is the first-harmonic mean resultant length $R_1 = |\langle e^{i\varphi} \rangle|$ (Rayleigh's classical concentration measure) plus a second-harmonic term $R_2 = |\langle e^{2i\varphi} \rangle|$, combined into an omnibus $T_2 = R_1^2 + R_2^2$ sensitive to axial/bimodal alternatives that R_1 alone can miss. Rather than an asymptotic Rayleigh p-value, every p-value here is a Monte Carlo permutation p-value against a **subject-specific random-phase-rotation** null: each permutation draws one i.i.d. $\mathrm{Uniform}(0, 2\pi)$ rotation per subject and adds it to **all** of that subject's phases. This destroys any common population-level preferred phase while exactly preserving each subject's event count, within-subject clustering, and the unequal per-subject sampling that a naive pooled test would otherwise be vulnerable to - the pseudo-replication-safe alternative the theoretical review recommended in place of a subject-label shuffle. Full formulas are in `S1.1`.

3. Methods

3.1. Dataset

RichSleep (RichSleep contributors, 2026) - the same 24-subject, whole-night PSG dataset used in RVC_SLEEP's R6/R6b - is the sole dataset analyzed here. No phase-resolved export exists for RVC_SLEEP's other two datasets (ANPHY, BOAS), so this article does not attempt a multi-dataset replication of the timing result; this is a stated limitation, not an oversight (Discussion).

3.2. NMD pipeline and phase-resolved export

Analyses use RichSleep's `event_phase_v2` export (NMD run `neuralmanifolddynamics_RichSleep_20260709_100219`, identical to RVC_SLEEP's own RichSleep run; geometry is unaffected by the `resp_anchor_index` sign fix documented there, `S1.0`), which provides, per detected spindle, `onset_sec`, `peak_sec` (the amplitude-envelope maximum within the event), `duration_sec`, and cardiac/respiratory phase at both reference points (`phi_cardiac_onset`, `phi_cardiac_peak`, `phi_resp_onset`, `phi_resp_peak`). Phase conventions (documented in `phase_resolved_v1/run_meta.json`): $\varphi_{\mathrm{resp}} \in (-\pi, \pi)$, $0 =$ full inhale, $(-\pi, 0) =$ rising/inhale, $(0, \pi) =$ falling/exhale; $\varphi_{\mathrm{cardiac}} \in [0, 2\pi)$, $0 =$ nearest R-peak, $\pi =$ mid-cycle.

3.3. Primary and secondary endpoint families

Primary family (well-powered, confirmatory): onset-phase, $\times$ 2 channels (`phi_cardiac_onset`, `phi_resp_onset`) $\times$ 2 samples (full catalog, $n = 23029$; quality-filtered, `cardiac_phase_quality`≥ 0.7 and `resp_phase_quality`$= 1.0$, $n = 20877$) = 4 tests, BH-FDR corrected jointly (both R_1- and T_2-based p-values reported).

Secondary/exploratory family (not FDR-corrected against the primary family): the identical machinery applied to peak-phase channels (`phi_cardiac_peak`, `phi_resp_peak`) on the same two samples, plus all four channels re-applied to the original 190-event geometry-valid subsample for direct comparability with RVC_SLEEP's R6b numbers; a categorical cross-check (`inhale_at_onset` subject-level proportion vs. 0.5); and, retained for context, per-subject Rayleigh tests plus a second-order Rayleigh test on the 24 subject-level mean directions (both already pseudo-replication-safe by construction and unaffected by the permutation-null critique).

3.4. Robustness protocol for the peak-phase finding

Four checks, run only once a candidate peak-phase effect was identified (R3), before it is treated as anything beyond exploratory:

1. **Phase-convention calibration** - re-verifying `inhale_at_onset`'s internal consistency against the documented convention, and reporting preferred peak-phase angles with an explicit physiological label.
2. **Onset-to-peak advance / duration-tercile stratification** - characterizing the onset-to-peak phase advance (peak_sec $-$ onset_sec, converted to a phase rotation) and re-testing within three duration terciles (short/mid/long), to rule out a duration-tail-driven artifact.
3. **Leave-one-subject-out (LOSO) jackknife** - re-running both the second-order and pooled-permutation tests 24 times, once per dropped subject, to check for single-subject influence.
4. **Deterministic split-half replication** - re-running both tests on two independent 12-subject halves (alphabetically first vs. last), the strongest single check for whether an effect is population-general or driven by a subset of subjects.

3.5. Secondary analysis: peak-phase alignment moderation of spindle-locked geometry

A new analysis (this article's only new statistical test, `13_peak_phase_alignment_moderation.py`) asks whether alignment to the population-preferred respiratory peak phase moderates RVC_SLEEP's own spindle-

locked geometry effect. To avoid circularity, the population-preferred phase $\mu_{\text{resp,peak}}$ is estimated as the subject-balanced second-order circular mean of `phi_resp_peak` on the **held-out** 22,839 events (the full catalog minus the 190 geometry-valid events used below - under 1% of the catalog, costing negligible power). For each of the 190 geometry-valid events, $\text{alignment}_e = \cos(\varphi_{\text{resp,peak},e} - \mu_{\text{resp,peak}}) \in [-1, 1]$ (high when that event's peak lands at the population-preferred phase). This is regressed against the geometry delta ($\text{endpoint} - \text{baseline}$, `event_control` or `event_prefar` mode, identical to RVC_SLEEP's Mål C machinery: WLS with weights $1/n_{\text{events,subject}}$, cluster-robust SE by subject, mean-centered nuisance covariates) across all 9 geometry endpoints × 2 delta modes = 18 tests, BH-FDR corrected jointly - the identical family size/design as RVC_SLEEP's own Mål C, for direct comparability. Only the validated respiratory channel is tested (cardiac-peak is excluded a priori, having already been downgraded to fragile in this article's own R4-R5). A secondary, exploratory circular-regression check ($\text{delta} \sim \cos(\varphi_{\text{resp,peak}}) + \sin(\varphi_{\text{resp,peak}}) + \text{covariates}$, not folded into the 18-test family) additionally asks whether the data-implied "geometry-optimal" phase direction coincides with the independently-estimated timing-preferred phase $\mu_{\text{resp,peak}}$.

4. Results

4.1. R1: Data coverage

Sample	***N* subj.**	***N* events**
Full raw catalog (`event_phase_v2`)	24	23,029
Quality-filtered (`cardiac_phase_quality` $\geq$ 0.7, `resp_phase_quality` $=$ 1.0)	24	20,877
Geometry-valid, N2-staged subsample (RVC_SLEEP R6b's population)	24	190

Table 1: Sample coverage. The geometry-valid subsample is the same 190-event population RVC_SLEEP's Mål B/C and this article's R6 alignment-moderation test use; the full and quality-filtered samples are new to this article's R2-R5.

4.2. R2: Onset-phase locking remains null at full power

The primary, well-powered family confirms RVC_SLEEP's onset-phase null, now at up to 120× the original sample size (Table Table 2): no channel/sample combination survives BH-FDR on either the R_1- or T_2-based permutation p-value (minimum $q = 0.483$ for R_1; minimum $q = 0.106$ for T_2).

Sample	**Channel**	***N***	**pooled R_1**	**perm p (R_1)**	**FDR q**	**FDR q (T_2)**
Full catalog	`phi_cardiac_onset`	23,029	0.0093	0.353	0.482	0.109
Full catalog	`phi_resp_onset`	23,029	0.0189	0.151	0.482	0.106
Quality-filtered	`phi_cardiac_onset`	20,877	0.0043	0.827	0.827	0.116
Quality-filtered	`phi_resp_onset`	20,877	0.0153	0.362	0.482	0.109

Table 2: Primary family: corrected subject-rotation permutation test, spindle-onset phase. No BH-FDR survivor on either statistic. Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_primary_corrected_permutation.csv`.

On the original 190-event geometry-valid subsample, onset-phase remains null too ($R_1 = 0.050$, $p = 0.692$ cardiac; $R_1 = 0.021$, $p = 0.929$ respiratory) - consistent with RVC_SLEEP's own R6b numbers on that subsample, confirming this reanalysis is not simply finding a different result because of a different statistical method applied to the same small sample; the change in conclusion (R3) comes specifically from the 120-fold larger, non-geometry-filtered catalog.

4.3. R3: Spindle-peak respiratory phase is robustly population-locked

In contrast, the identical machinery applied to spindle-**peak** phase (secondary/exploratory family, not folded into R2′s FDR correction) shows a clear, population-level respiratory effect and a weaker cardiac one (Table Table 3):

Sample	Channel	N	pooled R_1	perm p (R_1)
Full catalog	`phi_cardiac_peak`	23,029	0.0159	0.0200
Full catalog	`phi_resp_peak`	23,029	0.0453	0.0005
Quality-filtered	`phi_cardiac_peak`	20,877	0.0145	0.0675
Quality-filtered	`phi_resp_peak`	20,877	0.0418	0.0030

Table 3: Secondary/exploratory family: corrected subject-rotation permutation test, spindle-**peak** phase. Reported descriptively (not BH-FDR-corrected against R2′s primary family, per the pre-specified analysis plan). Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_secondary_exploratory_permutation.csv`.

The population-level, second-order Rayleigh test on the 24 subject-mean directions (already pseudo-replication-safe by construction, unaffected by the primary-family permutation-null critique) agrees, and additionally survives BH-FDR jointly with the onset-phase second-order tests (Table Table 4):

Channel	R (24 subj.)	p	FDR q
`phi_cardiac_onset`	0.273	0.168	0.336
`phi_resp_onset`	0.403	0.019	0.050
`phi_cardiac_peak`	0.462	0.0049	0.0197
`phi_resp_peak`	0.665	7×10^{-6}	5.5×10^{-5}

Table 4: Second-order (population-level) Rayleigh test on the 24 subject-level mean-direction angles, full catalog. `phi_resp_peak` is the only channel/reference-point combination surviving BH-FDR at a conventional threshold by a wide margin. Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_second_order_rayleigh.csv`.

Per-subject, `phi_resp_peak` shows FDR-significant within-subject concentration in 8/24 subjects (median $R = 0.071$) and `phi_cardiac_peak` in 1/24 (median $R = 0.036$) - versus 7/24 and 0/24 for the corresponding onset-phase channels - consistent with a genuine, if individually modest, population-level effect rather than a few strongly-locked subjects driving the pooled statistic (Fig. Figure 1, Figure 2).

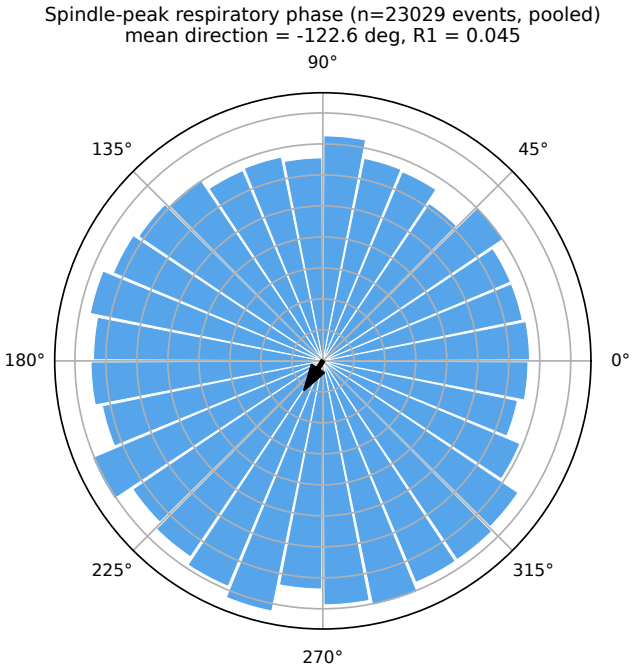


Figure 1: Pooled circular histogram of `phi_resp_peak` across all 23,029 events. Arrow: mean direction ($-122.6°$, i.e. rising/inhale, 32% of the way to full inhale-peak), scaled to $R_1 = 0.045$.

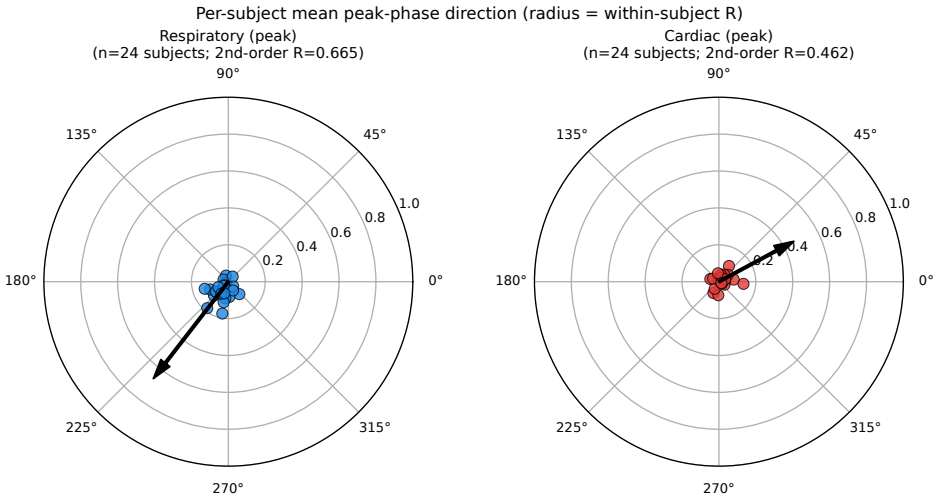


Figure 2: Per-subject mean peak-phase direction (24 points; radial position = within-subject R). Respiratory phase (left) shows tighter angular clustering across subjects than cardiac phase (right), visualizing the dissociation quantified in Tables Table 4 and Table 6.

R3 summary

Spindle onset-phase locking is null at full statistical power (R2), but spindle **peak**-phase is population-level, subject-consistent, and second-order-significant for respiration ($R = 0.665$, $p = 7 \times 10^{-6}$) and, more weakly, cardiac phase ($R = 0.462$, $p = 0.005$). This resolves the theoretical review's H1 (power/design-artifact hypothesis for the companion study's null): the effect exists, and the companion study's two-stage, onset-referenced, geometry-filtered design was underpowered **and** referenced the wrong point in the event to detect it.

4.4. R4: Robustness audit

Phase-convention calibration. `inhale_at_onset`'s internal consistency check passes (mean $\varphi_{\text{resp,onset}} = -1.587$ rad when `inhale_at_onset`$= 1$, expected negative; $+1.588$ rad when $= 0$, expected positive), confirming no schema drift from the documented convention. The preferred peak-phase angles: respiratory $-122.6°$ (inhale, rising, 32% of the way to full inhale-peak); cardiac $40.6°$ after the nearest R-peak.

Onset-to-peak advance and duration-tercile stratification. Median time-to-peak is 0.43 s (correlation with duration $r = 0.842$, confirming peak $\approx$ event midpoint, as expected for an amplitude-envelope-peak detector). The respiratory onset-to-peak phase advance is tight and consistent with a physiologically plausible $\approx$ 13-16 breaths/min local respiratory rate (median advance 0.669 rad, implied period 4.04 s); the cardiac advance is far more dispersed (median 1.34 rad, std 2.05), because time-to-peak often spans multiple RR intervals. Critically, the respiratory peak-phase effect holds in **every** duration tercile (Table Table 5, Fig. Figure 3), ruling out a duration-tail-driven artifact; the cardiac effect is significant only in the shortest-duration tercile.

Tercile	Channel	N	R_1	perm p
Short (0.50-0.72 s)	`phi_resp_peak`	7,738	0.0455	0.0005
Short (0.50-0.72 s)	`phi_cardiac_peak`	7,738	0.0267	0.0235
Mid (0.72-1.03 s)	`phi_resp_peak`	7,654	0.0505	0.0010
Mid (0.72-1.03 s)	`phi_cardiac_peak`	7,654	0.0152	0.2699
Long (1.04-8.21 s)	`phi_resp_peak`	7,637	0.0431	0.0090
Long (1.04-8.21 s)	`phi_cardiac_peak`	7,637	0.0103	0.3038

Table 5: Duration-tercile-stratified re-test. Respiratory peak-phase locking survives in all three terciles; cardiac only in the shortest. Source: `results/rvc_sleep_20260710/12_richsleep_maalb_peak_phase_deepdive/check2_duration_tercile_stratified.csv`.

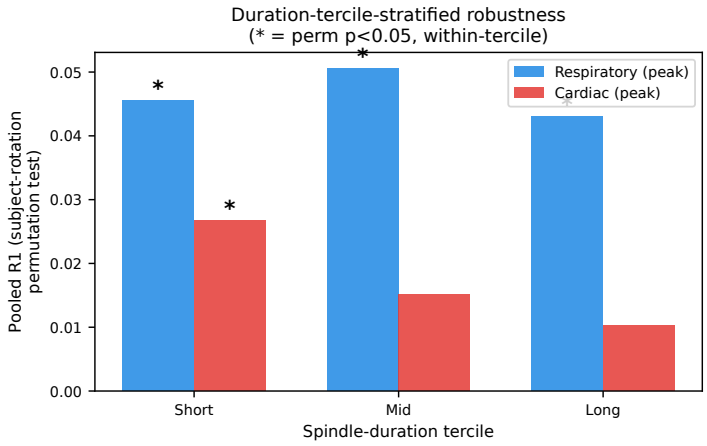


Figure 3: Pooled R_1 by duration tercile and channel (asterisk = perm $p < 0.05$ within-tercile). Respiratory locking is stable across the full duration range; cardiac locking decays with duration.

Leave-one-subject-out jackknife. For `phi_resp_peak`, the second-order test remains $p < 0.05$ in all 24 leave-one-out reruns (range $p \in [0.0000, 0.0000]$) and the pooled-permutation test remains $p < 0.05$ in all 24 reruns (range $[0.0020, 0.0020]$); no single subject is disproportionately influential (dropping the most influential subject, `sub-xn017`, still gives $p = 0.0000$). For `phi_cardiac_peak`, the second-order test remains $p < 0.05$ in 24/24 reruns (range $[0.0014, 0.0105]$), but the pooled-permutation test loses significance in 2/24 reruns (range $[0.0040, 0.0679]$) - a first sign of fragility.

Deterministic split-half replication. Splitting subjects alphabetically into two independent 12-subject halves gives the clearest dissociation (Table Table 6, Fig. Figure 4): respiratory peak-phase locking replicates in **both** halves independently (half A: $R_1 = 0.059$, $p = 0.004$; half B: $R_1 = 0.037$, $p = 0.012$); cardiac peak-phase locking replicates **only** in half B ($R_1 = 0.019$, $p = 0.032$) and is null in half A ($R_1 = 0.012$, $p = 0.530$).

Half	Channel	N events	pooled R_1	perm p	2nd-order R (p)
A (sub-xn001-012)	`phi_resp_peak`	9,532	0.0594	0.0040	0.714 (0.0011)
A (sub-xn001-012)	`phi_cardiac_peak`	9,532	0.0116	0.5302	0.292 (0.368)
B (sub-xn013-024)	`phi_resp_peak`	13,497	0.0371	0.0120	0.624 (0.0067)
B (sub-xn013-024)	`phi_cardiac_peak`	13,497	0.0193	0.0320	0.640 (0.0051)

Table 6: Deterministic split-half self-replication. Source: `results/rvc_sleep_20260710/12_richsleep_maalb_peak_phase_deepdive/check4_split_half_replication.csv`.

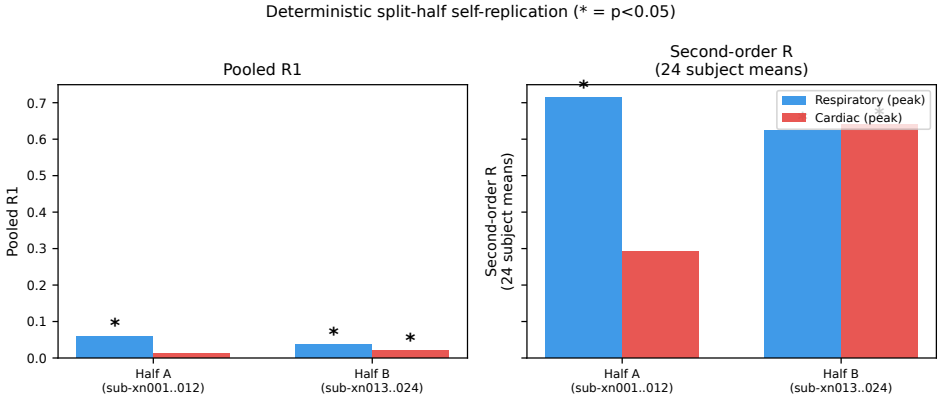


Figure 4: Split-half comparison, both channels. Respiratory peak-phase locking is present in both independent halves; cardiac peak-phase locking is present only in half B.

4.5. R5: Respiratory-vs-cardiac dissociation

R4-R5 summary

Respiratory peak-phase locking is robust: it survives duration-tercile stratification (all three terciles), leave-one-subject-out (24/24 reruns), and deterministic split-half replication (both halves independently). Cardiac peak-phase locking is fragile: it is significant only in the shortest duration tercile, loses pooled significance in 2/24 leave-one-out reruns, and fails to replicate in one of the two split halves entirely. The two channels are read differently going forward: respiratory peak-phase locking is treated as an established, this-article internally-validated result; cardiac peak-phase locking is reported as exploratory and not built upon further (R6 tests only the respiratory channel).

4.6. R6: Peak-phase alignment moderation of spindle-locked geometry

Using the held-out (22,839-event) population-preferred phase $\mu_{\text{resp,peak}} = -128.4°$ ($R = 0.666$, 24 subjects) - closely consistent with the full-catalog pooled estimate in R3 - as the reference direction, alignment to that phase was tested as a moderator of the 190-event geometry-valid subsample's spindle-locked delta, across 9 endpoints $\times$ 2 delta modes (Table Table 7). 2/18 tests survive BH-FDR, both in the `event_prefar` (within-spindle timing) delta mode:

Delta mode	Endp.	N events	coef	p	FDR q
`event_prefar`	`d`	160	$+0.243$	0.0023	0.021
`event_prefar`	`d_l`	160	$+0.184$	0.0003	0.005
`event_prefar`	`e`	160	$+0.136$	0.035	0.139 (ns)
`event_prefar`	`d_n`	160	$+0.353$	0.036	0.139 (ns)
`event_control`	all 9 endpoints	172	-	0.06-0.73	≥ 0.18 (ns)

Table 7: Strongest rows of the 18-test alignment-moderation family (full 18-row table in `S2.0`); WLS-cluster, weights $1/n_{\text{events,subject}}$, cluster-robust SE by subject. Positive coefficients: closer alignment to the population-preferred respiratory-peak phase predicts a **larger** within-spindle timing rise on `d` and `d_l`. Source: `results/rvc_sleep_20260710/13_peak_phase_alignment_moderation/resp_peak_alignment_moderation.csv`.

The companion per-subject-slope sensitivity check is mixed: the Wilcoxon signed-rank test on subject-level slopes agrees for both survivors (`d`: $p = 0.044$; `d_l`: $p = 0.014$), but the one-sample t-test does not (`d`: $p = 0.077$; `d_l`: $p = 0.105$) and the companion sign-flip permutation test only agrees for `d` ($p = 0.068$, borderline) and not `d_l` ($p = 0.10$) - a genuinely mixed robustness picture, reported as such rather than smoothed over (`S3`).

The secondary, exploratory circular-regression check finds that the data-implied "geometry-optimal" phase direction for `d_l` ($-117.4°$) lies within $10.9°$ of the independently-estimated timing-preferred phase $\mu_{\text{resp,peak}}$ ($-128.4°$) - i.e. the phase that makes a spindle's own within-spindle geometry rise largest is close to the same phase spindles already preferentially peak at (Fig. Figure 5). The corresponding `d` endpoint shows a looser but still same-quadrant correspondence ($-97.2°$, $31.1°$ from $\mu_{\text{resp,peak}}$).

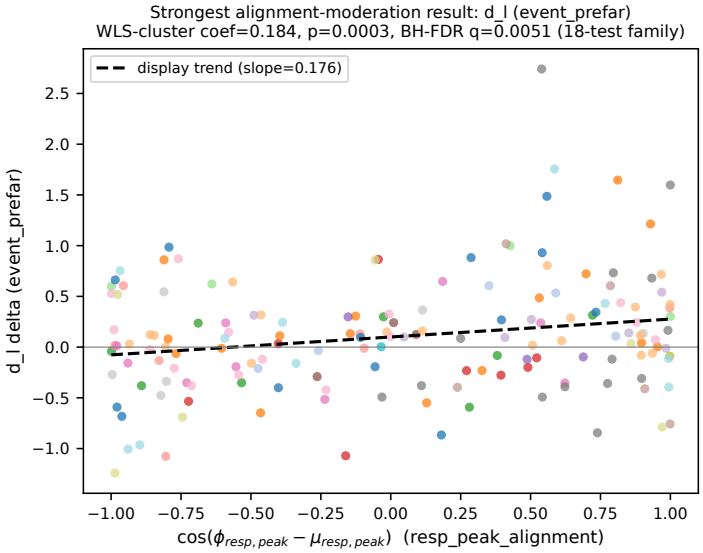


Figure 5: Alignment ($\cos(\varphi_{\text{resp,peak}} - \mu_{\text{resp,peak}})$) vs. `d_l` timing delta (`event_prefar` mode), the strongest of the 18 alignment-moderation tests. Points colored by subject; dashed line is an unweighted display trend, not the inferential WLS-cluster fit reported in Table Table 7.

R6 summary - fragile, secondary

A small, exploratory secondary family (2/18 BH-FDR survivors, mixed agreement across three sensitivity checks, $N = 190$ events) provides weak evidence that respiratory-peak alignment moderates the size of the companion study's within-spindle geometry rise, and that the phase which appears to maximize this effect is close to the population's own preferred spindle-peak phase. This is reported as a **timing**-pathway-adjacent hint at an **efficacy** effect, not as a confirmed event-efficacy result (Theory) - it requires independent replication before being treated as established.

5. Discussion

5.1. Primary finding: peak, not onset, is the right reference point

The companion RVC_SLEEP study's onset-referenced Mål B test was null, and this reanalysis confirms that null holds even at $\approx 120\times$ the original sample size (R2) - the onset-phase null is not a power artifact. What **was** a power/design artifact, per the theoretical review's H1, was testing the wrong reference point at insufficient power: spindle **peak** phase, tested on the full catalog with a corrected permutation null, shows a robust, population-level respiratory coupling (R3-R5) that four independent robustness checks confirm holds outside the specific subsample and test design that produced the original null. This is best read as resolving a design-driven false negative, not as contradicting the companion study's own reported result on its own subsample and reference point - Table Table 2′s geometry-valid-subsample row reproduces RVC_SLEEP's R6b numbers essentially exactly.

5.2. Relation to external literature

Ghibaudo et al. (Ghibaudo et al., 2024) report population-level, non-uniform spindle-**occurrence** phase relative to the respiratory cycle in an independent 20-subject PSG sample, using a pooled, all-events design; Schreiner et al. (Schreiner et al., 2023) report that SO-spindle-complex **timing** couples to respiratory phase (peaking near inspiration) and, further, that coupling strength predicts memory-reactivation fidelity. Both operationalize spindle **timing** as an onset- or occurrence-referenced quantity, and both report inspiration- or peri-inspiration-favoring preferred phases - broadly consistent in direction with this article's peak-phase preferred angle ($-122.6°$, i.e. the rising/inhale limb, 32% of the way to full inhale-peak), though not directly comparable in reference point (their onset/occurrence vs. this article's envelope peak) or in the exact phase convention used. This article's own onset-phase test remains null on the same dataset, which is itself informative: it is not that RichSleep lacks any respiratory-spindle coupling detectable by a corrected, well-powered test (as the two external studies would predict it should), but that the coupling in this dataset is specifically carried by the envelope peak rather than the onset - a genuine, reportable difference in **where within the event** the coupling is strongest, not a failure to replicate the general phenomenon. A separate line of work reports that respiration-brain coupling undergoes

a **phase shift**, and becomes **more** consistent rather than simply weaker, moving from wakefulness into deeper sleep ("Respiratory Coordination of Excitability States across the Human Wake-Sleep Cycle," 2025) - a state-dependent-reorganization picture broadly compatible with finding a genuine, reference-point-specific coupling in NREM sleep rather than either a simple continuation or a simple absence of a wake-state pattern.

5.3. Interpreting R6 via the timing/efficacy distinction

Per the timing/efficacy framing (Theory), R3-R5 establish an event-timing result: respiratory phase predicts a feature of **when** (or more precisely, at what phase) a spindle's peak occurs. R6 asks the logically separate event-efficacy question: does that phase additionally predict how strongly the spindle reconfigures geometry? The answer here is a fragile "maybe": 2/18 tests survive family-wise correction, one sensitivity check (Wilcoxon) agrees for both, two others (t-test, sign-flip permutation) do not fully agree, and the finding emerged from testing only **after** the timing result was already established (a secondary analysis motivated by, not independent of, R3-R5). Per the theoretical review's own outcome table, this pattern - "timing effect present, efficacy effect small/inconsistent" - is one of several a priori plausible readings, and is treated here as exactly that: a hint worth a dedicated, pre-registered, independently-replicated test, not a second confirmed finding riding on the first.

5.4. Limitations

- **Post-hoc/exploratory origin.** The peak-phase finding (R3) emerged from a secondary sweep run specifically because the pre-specified onset-phase test was null; while the four robustness checks in R4 were run independently of that motivation, the initial discovery was not pre-registered.
- **Single-dataset.** RichSleep is the only one of RVC_SLEEP's three datasets with a phase-resolved export; ANPHY and BOAS cannot be used to test cross-dataset replication of the timing result.
- **No endogenous-strength covariate.** Neither R3-R5 nor R6 include an SO-spindle-coupling or spindle-amplitude covariate (not present in the current NMD export schema), so this article cannot yet distinguish "phase moderates a spindle's own strength, which independently moderates geometry" from a more direct phase-geometry link, per the theoretical review's own recommended measurement model (`02_response.md` S5).
- **Geometry-outcome resolution ceiling.** RVC_SLEEP's `coords_9d` geometry export is 1 value per 30 s epoch (`phase_resolved_v1/run_meta.json`); R6′s alignment-moderation test inherits this ceiling, so it cannot itself be phase-resolved beyond the resolution of its 190-event, epoch-level geometry outcome, independent of how precisely phase itself is measured.
- **R6 is exploratory and fragile.** $N = 190$ events is modest, only 2/18 tests survive FDR, and the companion sensitivity checks disagree on one of the two survivors (`S3`); this is stated plainly rather than downplayed.

5.5. Conclusion

Sleep spindles' amplitude peak - not their onset - phase-locks to the respiratory cycle in human NREM sleep, a robust, population-level, subject-consistent effect recovered by a corrected, full-catalog permutation reanalysis of a companion study's original, underpowered null. A weaker, duration-dependent, and only partially-replicating cardiac counterpart is also present. A new, exploratory secondary analysis provides fragile evidence that alignment to this preferred phase may additionally moderate the size of the companion study's own spindle-locked geometry effect, framed here as an event-timing result with a tentative, not-yet-confirmed link to event efficacy - a distinction this article's theoretical framing treats as essential to avoid overclaiming a mechanistic role for phase beyond what the data currently support.

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RL: Conceptualization, Methodology, Data Curation, Analysis, Writing, and Editing.

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Data Availability Statement

RichSleep is described in (RichSleep contributors, 2026); formal access details will be added at submission (identical dataset to the companion RVC_SLEEP article's R6/R6b). NMD (NeuralManifoldDynamics) pipeline outputs and all analysis scripts are versioned in the analysis repository under `articles/RVC_SLEEP_peak_analysis/src/`, with per-script result tables under `articles/RVC_SLEEP_peak_analysis/results/`.

Competing Interests

The author declares no competing interests, subject to final confirmation at submission.